A logarithm over the cyclotomic quadratic

one parameter, one partial fraction, and a rational function that collapses

$$\boxed{\int_0^\infty \frac{\log(1+x^3)}{1+x+x^2}\,\mathrm{d}x = \frac{2\pi\log 2}{\sqrt{3}}.}$$

Overview. Introduce the parameter

$$F(t) = \int_0^\infty \frac{\log(1+t^3x^3)}{1+x+x^2}\,\mathrm{d}x, \qquad t \ge 0,$$

so that the problem is $F(1)$. Differentiating in t turns the logarithm into a rational function, and that rational integral has a simple value: the whole cubic–quadratic interaction cancels and leaves

$$\int_0^\infty \frac{\mathrm{d}x}{(1+t^3x^3)(1+x+x^2)} = \frac{2\pi}{3\sqrt{3}} \cdot \frac{1}{1+t}.$$

Hence $F'(t) = \frac{2\pi}{\sqrt{3}} \cdot \frac{1}{1+t}$, and integrating back from $F(0) = 0$ gives

$$\boxed{F(t) = \frac{2\pi}{\sqrt{3}}\log(1+t), \qquad t \ge 0,}$$

of which the stated integral is the case $t = 1$.

The reason the constant $\frac{2\pi}{3\sqrt{3}}$ governs everything is that $\frac{\mathrm{d}x}{1+x+x^2}$ is, up to the factor $\frac{1}{\sqrt{3}}$, arclength on a circular arc; Section 5 makes this precise.

1. The parametrised integral

For $t > 0$ put

$$F(t) = \int_0^\infty \frac{\log(1+t^3x^3)}{1+x+x^2}\,\mathrm{d}x.$$

Convergence is immediate: at 0 the integrand vanishes like t^3x^3, and at infinity it behaves like $\frac{3\log x}{x^2}$. Also $F(0) = 0$.

Differentiating under the integral sign,

$$F'(t) = \int_0^\infty \frac{3t^2x^3}{(1+t^3x^3)(1+x+x^2)}\,\mathrm{d}x. \tag{1}$$

The differentiation is legitimate on any interval $0 < t_1 \le t \le t_2$: the t-derivative of the integrand is bounded by $\frac{3}{t_1} \cdot \frac{1}{1+x+x^2}$, because $\frac{t^3x^3}{1+t^3x^3} \le 1$, and that bound is integrable.

The identity

$$3t^2x^3 = \frac{3}{t}\big[(1+t^3x^3) - 1\big]$$

splits (1) into

$$F'(t) = \frac{3}{t}\left[\int_0^\infty \frac{\mathrm{d}x}{1+x+x^2} - \int_0^\infty \frac{\mathrm{d}x}{(1+t^3x^3)(1+x+x^2)}\right] = \frac{3}{t}\big[C - K(t)\big], \tag{2}$$

where

$$C = \int_0^\infty \frac{\mathrm{d}x}{1+x+x^2}, \qquad K(t) = \int_0^\infty \frac{\mathrm{d}x}{(1+t^3x^3)(1+x+x^2)}.$$

2. The base constant

Completing the square,

$$C = \int_0^\infty \frac{\mathrm{d}x}{\left(x+\frac{1}{2}\right)^2+\frac{3}{4}} = \frac{2}{\sqrt{3}}\left[\arctan\frac{2x+1}{\sqrt{3}}\right]_0^\infty = \frac{2}{\sqrt{3}}\left(\frac{\pi}{2}-\frac{\pi}{6}\right),$$

so

$$\boxed{C = \int_0^\infty \frac{\mathrm{d}x}{1+x+x^2} = \frac{2\pi}{3\sqrt{3}}.} \tag{3}$$

3. The rational lemma

Lemma. For every $t>0$,

$$\boxed{K(t) = \int_0^\infty \frac{\mathrm{d}x}{(1+t^3x^3)(1+x+x^2)} = \frac{2\pi}{3\sqrt{3}}\cdot\frac{1}{1+t} = \frac{C}{1+t}.} \tag{4}$$

Proof. Write $P(x)=1+t^3x^3$ and $Q(x)=1+x+x^2$, and seek the partial fraction decomposition

$$\frac{1}{P(x)Q(x)} = \frac{\alpha(x)}{P(x)} + \frac{\beta(x)}{Q(x)}, \qquad \deg\alpha \le 2,\ \deg\beta \le 1.$$

Equivalently $1 = \alpha Q + \beta P$. Evaluate at the roots of Q, namely the primitive cube roots of unity $\omega = \mathrm{e}^{2\pi\mathrm{i}/3}$ and $\overline{\omega}$. Since $\omega^3 = \overline{\omega}^3 = 1$,

$$P(\omega) = 1+t^3\omega^3 = 1+t^3 = P(\overline{\omega}),$$

so

$$\beta(\omega) = \beta(\overline{\omega}) = \frac{1}{1+t^3}.$$

A polynomial of degree ≤ 1 taking equal values at two distinct points is constant, hence

$$\beta(x) \equiv \frac{1}{1+t^3}. \tag{5}$$

This is the only place where the cubic structure is used, and it is exactly the statement that Q divides x^3-1.

For α, subtract:

$$\frac{\alpha(x)}{P} = \frac{1}{PQ} - \frac{1}{(1+t^3)Q} = \frac{(1+t^3)-P(x)}{(1+t^3)PQ} = \frac{t^3\left(1-x^3\right)}{(1+t^3)PQ}.$$

Now $1-x^3 = (1-x)(1+x+x^2) = (1-x)Q$, so the factor Q cancels and

$$\alpha(x) = \frac{t^3(1-x)}{1+t^3}. \tag{6}$$

Thus

$$\frac{1}{P(x)Q(x)} = \frac{1}{1+t^3}\left[\frac{t^3(1-x)}{1+t^3x^3} + \frac{1}{1+x+x^2}\right]. \tag{7}$$

Both pieces are separately integrable on $[0,\infty)$: the first decays like $-\frac{1}{t^3x^2}$.

For the first piece substitute $u = tx$:

$$\int_0^\infty \frac{1-x}{1+t^3x^3}\,\mathrm{d}x = \frac{1}{t}\int_0^\infty \frac{\mathrm{d}u}{1+u^3} - \frac{1}{t^2}\int_0^\infty \frac{u\,\mathrm{d}u}{1+u^3}.$$

Both integrals are instances of the beta-type formula

$$\int_0^\infty \frac{u^{s-1}}{1+u^3}\,\mathrm{d}u = \frac{\pi/3}{\sin\frac{\pi s}{3}}, \qquad 0 < s < 3, \tag{8}$$

and $\sin\frac{\pi}{3} = \sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$, so both equal $\frac{2\pi}{3\sqrt{3}} = C$. (Formula (8) follows from the standard $\int_0^\infty \frac{u^{s-1}}{1+u^n}\,\mathrm{d}u = \frac{\pi/n}{\sin(\pi s/n)}$; the coincidence of the two values here is the symmetry $s \mapsto 3 - s$.) Hence

$$\int_0^\infty \frac{1-x}{1+t^3x^3}\,\mathrm{d}x = C\left(\frac{1}{t} - \frac{1}{t^2}\right) = C \cdot \frac{t-1}{t^2}.$$

Substituting this and (3) into (7),

$$K(t) = \frac{1}{1+t^3}\left[t^3 \cdot C \cdot \frac{t-1}{t^2} + C\right] = \frac{C\big[t(t-1)+1\big]}{1+t^3} = \frac{C\,(t^2-t+1)}{(1+t)(t^2-t+1)} = \frac{C}{1+t},$$

using $1 + t^3 = (1+t)(t^2-t+1)$. □

Everything turns on the cancellation of $t^2 - t + 1$. The numerator $t(t-1)+1$ produced by the two beta integrals is literally the same cyclotomic factor that appears in the factorisation of $1+t^3$; nothing survives except the linear factor $1+t$, which is what makes the antiderivative a logarithm.

4. Conclusion

Insert (3) and (4) into (2):

$$F'(t) = \frac{3}{t}\left[C - \frac{C}{1+t}\right] = \frac{3C}{t} \cdot \frac{t}{1+t} = \frac{3C}{1+t} = \frac{2\pi}{\sqrt{3}} \cdot \frac{1}{1+t}. \tag{9}$$

Since $F(0) = 0$,

$$F(t) = \int_0^t \frac{2\pi}{\sqrt{3}}\frac{\mathrm{d}s}{1+s} = \frac{2\pi}{\sqrt{3}}\log(1+t),$$

that is,

$$\boxed{\int_0^\infty \frac{\log(1+t^3x^3)}{1+x+x^2}\,\mathrm{d}x = \frac{2\pi}{\sqrt{3}}\log(1+t), \qquad t \geq 0.} \tag{10}$$

Setting $t = 1$,

$$\int_0^\infty \frac{\log(1+x^3)}{1+x+x^2}\,\mathrm{d}x = \frac{2\pi\log 2}{\sqrt{3}} = 2.5144598308\ldots,$$

in agreement with numerical quadrature to the displayed precision.

Two consistency checks on (10). As $t \to 0^+$, the left side is $\approx t^3 \int_0^\infty \frac{x^3}{1+x+x^2}\,\mathrm{d}x$, which diverges, while the right side is $\approx \frac{2\pi t}{\sqrt{3}}$. There is no contradiction, because for small t the logarithm is only *eventually* small: the crossover happens at $x \sim 1/t$, and it is the tail beyond it that produces the linear behaviour. As $t \to \infty$, $\log(1+t^3x^3) \approx 3\log t + 3\log x$ and $3C\log t = \frac{2\pi}{\sqrt{3}}\log t$ matches the leading term of $\frac{2\pi}{\sqrt{3}}\log(1+t)$.

5. Why $2\pi/(3\sqrt{3})$: the arc behind the measure

The constant C is not arbitrary. Let $\omega = \mathrm{e}^{2\pi\mathrm{i}/3}$, so that $1 + x + x^2 = (x-\omega)(x-\overline{\omega})$, and set

$$z = \frac{x-\omega}{x-\overline{\omega}}.$$

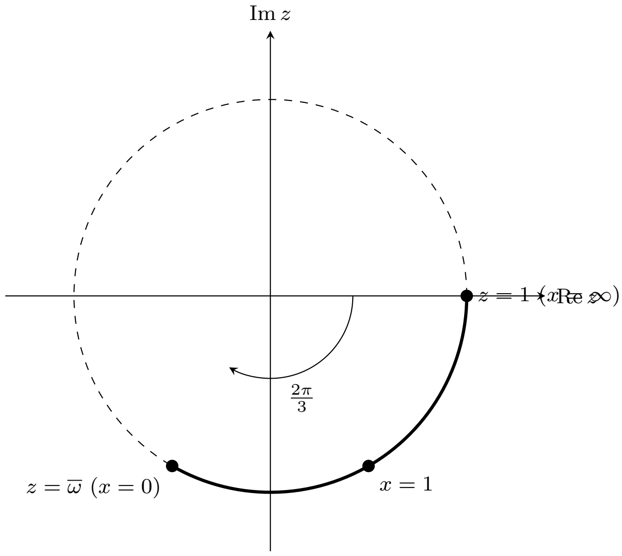
For real x the two distances $|x-\omega|$ and $|x-\overline{\omega}|$ are equal, so $|z|=1$: the real line is mapped to the unit circle. Moreover

$$\frac{\mathrm{d}z}{z}=\left(\frac{1}{x-\omega}-\frac{1}{x-\overline{\omega}}\right)\mathrm{d}x=\frac{\overline{\omega}-\omega}{(x-\omega)(x-\overline{\omega})}\,\mathrm{d}x=\frac{-\mathrm{i}\sqrt{3}}{1+x+x^2}\,\mathrm{d}x,$$

since $\omega-\overline{\omega}=\mathrm{i}\sqrt{3}$. Writing $z=\mathrm{e}^{\mathrm{i}\varphi}$, $\frac{\mathrm{d}z}{z}=\mathrm{i}\,\mathrm{d}\varphi$, so

$$\boxed{\frac{\mathrm{d}x}{1+x+x^2}=-\frac{\mathrm{d}\varphi}{\sqrt{3}}.} \tag{11}$$

The measure in the problem *is* arclength on the unit circle, scaled by $\frac{1}{\sqrt{3}}$.



total arc $\frac{2\pi}{3}$, hence $C=\frac{1}{\sqrt{3}}\cdot\frac{2\pi}{3}$

As x runs from 0 to ∞, z runs from $\frac{-\omega}{-\overline{\omega}}=\omega^2=\overline{\omega}$ to 1, covering an arc of angular length $\frac{2\pi}{3}$, exactly one third of the circle, because ω is a primitive cube root of unity. Therefore

$$C=\frac{1}{\sqrt{3}}\cdot\frac{2\pi}{3}=\frac{2\pi}{3\sqrt{3}},$$

recovering (3) with no antiderivative at all. The same picture explains why the answer to the whole problem carries a factor $\frac{\pi}{\sqrt{3}}$ rather than, say, π: the $\sqrt{3}$ is $|\omega-\overline{\omega}|$, and the $\frac{2\pi}{3}$ is the third of the circle cut out by the cyclotomic quadratic.

6. The pattern for other exponents

Nothing forces the exponent 3. The same three steps, writing $nt^{n-1}x^n=\frac{n}{t}\big[(1+t^nx^n)-1\big]$, decomposing, and using (8), apply whenever the denominator Q divides x^n-1, since that is what made β constant in (5).

$Q(x)$	divides	resulting identity
$1+x$	x^2-1	$\displaystyle\int_0^\infty \frac{\log(1+t^2x^2)}{1+x}\,\mathrm{d}x$ diverges; the quadratic Q is needed for convergence.
$1+x+x^2$	x^3-1	$\displaystyle\int_0^\infty \frac{\log(1+t^3x^3)}{1+x+x^2}\,\mathrm{d}x = \frac{2\pi}{\sqrt{3}}\log(1+t).$
$1+x^2$	x^4-1	$\displaystyle\int_0^\infty \frac{\log(1+t^4x^4)}{1+x^2}\,\mathrm{d}x = \pi\log(1+\sqrt{2}\,t+t^2)$, by the same route with $\int_0^\infty \frac{u^{s-1}}{1+u^4}\,\mathrm{d}u = \frac{\pi/4}{\sin(\pi s/4)}$.

In each case the mechanism is identical: the roots of Q are roots of unity, so P takes the same value $1+t^n$ at all of them, the correction term α inherits the factor Q from $1-x^n$, and the beta integrals supply a numerator that cancels the non-linear cyclotomic factor of $1+t^n$. What is left is always a logarithm.